UC San Diego
RADY SCHOOL OF MANAGEMENT

MGTA 415 NLP Applications for Unstructured Text Data

Summer 2024

INSTRUCTOR: Allison Chia-Yi Wu (alw050@ucsd.edu)

OFFICE HOURS: Fridays 10AM - 11AM

TEACHING ASSISTANT: Iman Sayyadzadeh

COURSE TIME: Saturdays 1-5PM

Overview

This course offers a comprehensive exploration of natural language processing (NLP) and large language models (LLM), with a focus on applying these technologies to real-world business problems. Over five weeks, we'll quickly review classical NLP techniques such as text preprocessing and embeddings, then move rapidly into advanced LLM applications, including transformers, prompt engineering, multimodal models, and agent-based AI interactions. This journey equips students with foundational skills and state-of-the-art methodologies needed to transform unstructured text into valuable business insights.

The course emphasizes high-level understanding of key NLP concepts through real-world applications, interactive demos, and hands-on lab sessions. Specifically, students will work extensively on one comprehensive group project, presenting their progress in a midterm session and delivering a final presentation at the end of the course.

By the end of this course, students will have a strong understanding of advanced NLP and LLM applications, enabling them to leverage these technologies effectively across diverse business scenarios, significantly enhancing their analytical capabilities and strategic impact.

About the Instructor

Allison Chia-Yi Wu, Ph.D., earned her doctorate in Bioinformatics from UCSD in 2016 and brings over eight years of industry experience in data science. Her expertise spans various domains, including bioinformatics, marketing, digital healthcare, and her latest venture into climate tech. Allison has held roles ranging from individual contributor to leadership positions, where she has built and led data science teams and developed robust data architectures. Currently, she serves as VP of Data Science at Nucleix, Inc., where she leads the data science department leveraging LLMs and ML in cancer diagnostics research. She is also the founder of TerraMinds.ai, an AI/ML consulting practice dedicated to delivering end-to-end, production-level AI/ML solutions across diverse domains.

Her specialty lies in end-to-end, production-level AI/ML implementations, encompassing everything from algorithm development to deployment. She is proficient in both classical machine learning algorithms and advanced NLP/LLM technologies. Allison's diverse experience includes working for large corporations like Thermo Fisher Scientific and pre-seed start-ups.

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Prerequisites

- Math, basic statistics
- Python Programming
- Basic knowledge of supervised machine learning

Schedule

Session	Date	Class Topic & Activities	Course Materials
Session 1	7/12	1. Lecture: NLP & LLM Fundamentals 2. Interactive Lab: Demo Conversation	Hugging Face NLP Course, NLTK Book Chapter 1, Transformers Explained
Session 2	7/26	1. Lecture: Prompt Engineering & Applications 2. Interactive Lab: AI Research Companion	Prompt Engineering Guide, Text Summarizer Demo
Session 3	8/9	1. Midterm Project Presentations 2. Lecture:LLM Agents & Tools Integration	LangChain Documentation, CrewAI Tutorial
Session 4	8/23	1.Lecture: Customizing LLMs with MCP & Retrieval-Augmented Generation 2.Interactive Lab: Conversational Evaluation (Agent Arena)	LangChain MCP Server Docs, Building a RAG Pipeline, Open LLM Leaderboard
Session 5	9/6	1. Lecture: Multimodal Models & Responsible AI 2. Final Project Presentations	Responsible AI Guide, Multimodal LLM Overview

Course Materials

- Lecture slides: Slides will be posted to Canvas on the evening before class
- Hugging Face NLP Course: https://huggingface.co/learn/nlp-course
- Natural Language Processing with Python: https://www.nltk.org/book/
- OpenAI APIs and Playground
- LangChain Documentation and Tutorials

Recommended Resources

- Hugging Face
- Prompt Engineering Guide
- LangChain
- Pinecone for Semantic Search
- FAISS Vector Store

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Group Project

Students will undertake one comprehensive group project throughout the course. Teams of 3 to 4 students should form at the start of the course. Each group will leverage techniques learned during the course—including classical NLP, sentiment analysis, and LLM-driven feature extraction—to develop an AI-driven application for one of the following 5 themes.

Project Themes

Project Theme	Goal	Suggested Data Sources
Robo Portfolio Advisor	Predictive financial analysis and portfolio management using LLMs.	Alpaca API, Financial Modeling Prep, SEC Edgar filings
Customer Intelligence Tool	Sentiment and insight analysis from customer feedback.	Amazon/Yelp/TripAdvisor reviews, Twitter API, Hugging Face datasets
AI Business Assistant	Internal company conversational AI for queries such as HR topics.	Enron Email Dataset, public corporate manuals, FAQ databases
Knowledge Base Extractor	Efficient querying of extensive documentation or knowledge bases.	Wikipedia API, public technical documentation or handbooks
Competitor Analysis Bot	Continuous competitor monitoring and market activity summarization.	NewsAPI, Google News, Twitter, Reddit feeds

Presentations

- **Midterm Project Progress (Session 3):** 10-minute presentation plus 10-minute Q&A session, highlighting project progress for feedback and guidance.
- **Final Project Presentation (Session 5):** 20-minute presentation plus 10-minute Q&A session, evaluated based on analytical reasoning, project comprehensiveness, and communication effectiveness.

Grading

- **Attendance and In-class participation: 30%**
    - **Attendance** (5%): You are expected to join each session, either remotely or in person. If you need to miss a class, you must email me and the TA to request approval at least 3 days before the lecture. **Missing more than one class will result in losing the entire attendance score (5%).**
    - **Quizzes & Discussions (10%)**: We will use iClicker for in-class quizzes and discussions. These will cover readings and materials from the previous session. Students are responsible for setting up their iClicker subscription (covered by Rady Business School).
    - **Interactive Lab (15%):** Graded based on engagement in small group discussions, contributions to in-lab problem-solving, and clarity in presenting group outcomes during lab share-outs.

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- **Midterm Group Project Presentation: 30%**
    - Graded based on performance during the midterm presentation and Q&A. Emphasis is on planning clarity, methodological reasoning, and team collaboration.
    - Grading guidelines:
        - **Clarity**: 10%
        - **Reasoning of the choice of analysis techniques**: 10%
        - **Collaborativeness**: 10% All team members must clearly articulate their contribution and understanding of the project.

- **Final Group Project Presentation: 40%**
    - Graded based on final presentation delivery and Q&A session. Emphasis is on depth, rigor, and clarity of communication.
    - Grading guidelines:
        - **Clarity**: 10%
        - **Comprehensiveness**: 10%
        - **Reasoning of the choice of analysis techniques**: 10%
        - Collaborativeness: 10% All team members must clearly articulate their contribution and understanding of the project.

COURSE POLICIES

Use of Generative AI

"AI should be used to augment human intelligence, helping us make better decisions and enhancing our capabilities." — Fei-Fei Li

In this course, you are expected to learn to work with generative AI. The baseline performance you will be benchmarked against is what can be generated by ChatGPT without advanced instructions. We will provide a baseline solution for all assignments, including the prompts used to generate these solutions.

If you choose not to use generative AI, you may find yourself at a disadvantage. This class is designed for you to collaborate with generative AI to create and explore solutions beyond the baseline, and to understand the reasoning behind different NLP techniques.

The course is structured to reward those who can effectively leverage generative AI to enhance their understanding of LLM/Gen AI technology. Your ability to use these tools to their fullest potential will be a key factor in your success.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

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The complete UCSD Policy on Integrity of Scholarship can be viewed at:
http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or **osd@ucsd.edu.**

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